SORA: Free Second-Order Attacks in Fast Adversarial Training

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Abstract

Adversarial Training (AT) is a leading defense against adversarial examples but often suffers from *Catastrophic Overfitting* (CO) in efficient single-step variants, where robustness to multi-step attacks collapses despite high single-step performance. We address this failure mode with two contributions. First, we formalize *Epsilon Overfitting* (EO), a perspective in which fixed perturbation magnitudes and directions exacerbate CO, and show that introducing perturbation variability significantly improves robust generalization across different architectures and datasets. Second, we propose **PertAlign** (Perturbation Alignment), a theoretically grounded, computationally negligible metric that predicts CO onset by measuring gradient alignment across attack stages. Leveraging these insights, we introduce **SORA**, an adaptive step-size AT method that dynamically adjusts perturbations based on loss surface geometry. SORA consistently prevents CO, achieves state-of-the-art robustness and clean accuracy, and generalizes across datasets and architectures using a single fixed set of hyperparameters, which is essential for applicability in fast AT. Extensive experiments on diverse datasets and architectures show that SORA matches or surpasses the robustness of prior methods while delivering higher clean accuracy and superior efficiency. Code is available at `https://github.com/SecondOrderAT/SORA`.

1. Introduction

Deep neural networks have achieved remarkable success across a wide range of domains. However, their vulnerability to adversarial examples poses significant challenges to their reliability and security (Szegedy et al., 2014). These adversarial examples are often subtle and imperceptible to the human eye, yet can cause models to make incorrect predictions with high confidence.

Adversarial Training (AT) (Madry et al., 2018) has emerged as one of the most effective defenses against adversarial examples (Athalye et al., 2018). By training models on adversarially perturbed samples, AT improves robustness to attacks during deployment. Despite these benefits, multi-step AT methods such as Projected Gradient Descent (PGD) (Madry et al., 2018) are computationally expensive, as they require several gradient ascent steps to generate adversarial examples at each training iteration. This cost severely limits their scalability to large datasets and deep architectures.

To mitigate this overhead, single-step adversarial training approaches (Wong et al., 2020) have been proposed. While significantly more efficient, these methods introduce new challenges. Notably, Wong et al. (2020) identified a failure mode in single-step AT, commonly observed when using the Fast Gradient Sign Method (FGSM) (Goodfellow et al., 2015), that has been termed *Catastrophic Overfitting* (CO); related limitations of one-step attacks were also discussed in earlier AT work (Madry et al., 2018). In this setting, robust accuracy against multi-step attacks such as PGD collapses to nearly 0%, while FGSM accuracy can rise sharply and even exceed the clean accuracy. This suggests that the model overfits to the specific attack used in training and becomes brittle against stronger iterative attacks.

Although many methods have been proposed to address this challenge, we argue that an effective fast AT method should satisfy the following properties:

- **Robustness across datasets and architectures.** It should reliably prevent CO on a diverse range of datasets and models. As shown in Section 6 and Appendix K, our proposed method, Second-Order Adaptive (SORA), succeeds not only on commonly used datasets but also on challenging benchmarks where existing single-step techniques often fail.
- **High clean and robust accuracy with low cost.** A known trade-off exists between clean and robust accuracy (Tsipras et al., 2019; Zhang et al., 2019b). Many prior methods achieve strong robustness by restrict-

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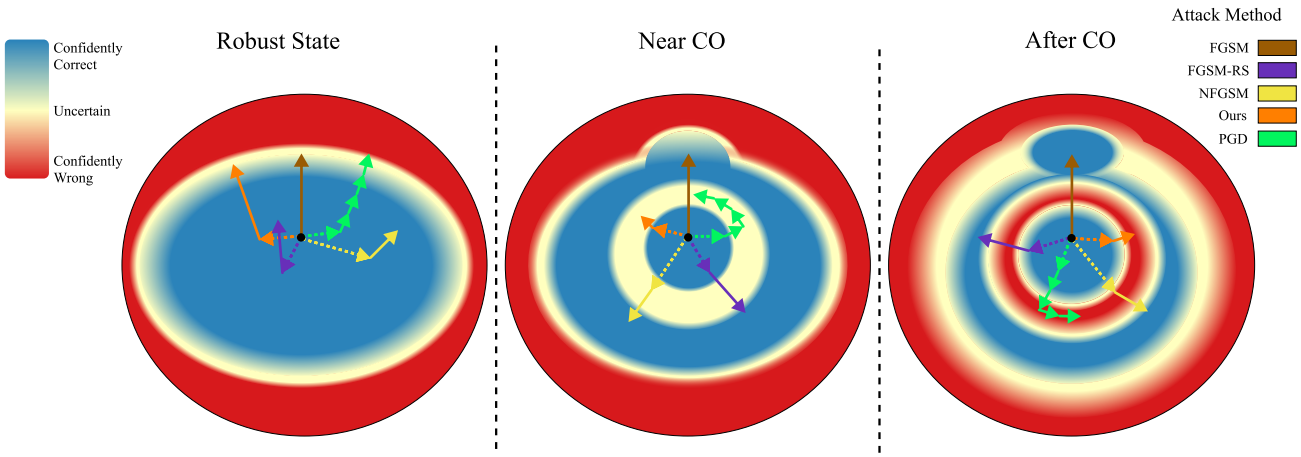


Figure 1. Evolution of the loss landscape geometry around a sample during FGSM training. **Left:** Early in training, before CO, the model is robust, with FGSM and PGD accuracies closely matched. The loss surface is approximately semilinear. **Middle:** A few batch updates before CO, the decision boundary begins to wrap around the original and adversarial examples, forming a nonlinear region that is not yet misclassified. **Right:** This nonlinearity escalates, leading to misclassification in that region, which stronger attacks such as PGD exploit. Perturbations from other attacks and our method (SORA) are also visualized.

ing model flexibility (e.g., stronger regularization or reduced capacity), which harms clean accuracy. In contrast, SORA maintains high clean accuracy and state-of-the-art robustness while adding negligible computational and memory overhead by efficiently choosing the attack step-size distribution.

- **Dataset and architecture agnostic hyperparameters.** Several methods require extensive, dataset- and model-specific hyperparameter tuning to avoid CO (Appendix H), which undermines the original goal of Fast Adversarial Training (FAT): robustness with minimal training cost. In our experiments (Section 6 and Appendix K), we fix SORA's hyperparameters across all settings and adopt the single best configuration reported in prior work for a fair comparison, demonstrating strong generalization without costly tuning for each setting.

To the best of our knowledge, no metric has been explicitly designed to predict CO without significant cost. Nevertheless, several existing measures can serve as indirect early indicators of CO onset (Andriushchenko & Flammarion, 2020; Lin et al., 2023; Rocamora et al., 2024). These indicators often incur high computational cost, requiring extra backpropagation or forward passes. To address this gap, we propose PertAlign, a fast, gradient-based metric that is theoretically linked to the curvature of the loss landscape and can predict CO with negligible overhead. It uses gradients from the first and second attack iterations. We further show that PertAlign can anticipate CO earlier than existing indicators.

In addition, we identify a previously overlooked phenomenon, which we term *Epsilon Overfitting* (Figure 1). We show in Section 3 that variability in the attack step-size plays a crucial role in the emergence of CO. By incorporating this insight, SORA achieves improved robustness across a wider range of settings compared to previous baselines. Our experiments demonstrate that methods incorporating perturbation variability generalize more effectively and consistently avoid CO.

Our contributions are as follows:

- We introduce and analyze *Epsilon Overfitting*, demonstrating its importance for robust generalization and CO.
- We propose *PertAlign*, a theoretically grounded and computationally efficient metric for early and reliable CO prediction.
- We present *SORA*, an efficient training paradigm that leverages these insights to adaptively determine attack step-sizes. We show that it matches or surpasses state-of-the-art generalization across datasets and architectures with minimal hyperparameter tuning.
- We identify key criteria for making fast methods deployable in practice and evaluate them on well-known domains, as well as on important but less frequently tested medical domains.

2. Background and Related Work

2.1. Adversarial Training

Adversarial Training (AT) can be formulated as a min–max optimization problem, where the inner maximization seeks adversarial perturbations that maximize the training loss,

while the outer minimization updates model parameters to minimize the loss on these adversarial examples (Madry et al., 2018). Formally, this can be expressed as:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\delta\in\Delta} \mathcal{L}\left(f_\theta(x+\delta), y\right) \right], \tag{1}$$

where θ denotes the parameters of the model f, (x, y) are the training data drawn from the distribution $\mathcal{D}$, $\mathcal{L}(\cdot, \cdot)$ denotes the cross-entropy loss, and δ is the perturbation confined within a given boundary Δ.

In the standard multi-step AT by (Madry et al., 2018), the loss function used is the cross-entropy loss and Δ is defined as the ℓ_∞-norm ball with radius ϵ, and the inner maximization is solved via the Projected Gradient Descent (PGD) attack. While effective, this iterative approach significantly increases the computational cost for large-scale models and datasets. Zhang et al. (2019b) use a regularization term that minimizes the Kullback–Leibler (KL) divergence between the output distributions of clean and adversarial examples.

Shafahi et al. (2019) use the gradient used for generating adversarial examples for updating the model weights at the same time and therefore eliminate the overhead of the attack.

Following Wong et al. (2020), a family of methods, collectively referred to as Fast Adversarial Training (FAT), emerged with the goal of producing robust models at a fraction of the cost of multi-step AT. However, while these methods improve efficiency, fast single-step AT approaches are prone to a distinctive failure mode known as Catastrophic Overfitting (CO). Understanding and mitigating CO has thus become a critical research direction.

2.2. Catastrophic Overfitting

CO is a phenomenon observed during AT, more prevalently during single-step AT, where robustness against multi-step attacks such as PGD suddenly decreases to 0% just within a few epochs, whereas robustness against the FGSM attack rapidly increases. Many studies have focused on explaining CO and developing FAT variants that avoid it. Broadly, prior works can be grouped by their main mitigation strategy:

Randomization. Randomization (noise-based) methods inject stochasticity into adversarial example generation to prevent harmful overfitting. Wong et al. (2020) demonstrated that adding random starts to FGSM training can prevent CO. de Jorge Aranda et al. (2022) extended this by increasing the magnitude of random noise and removing the clipping step to boost accuracy.

Regularization. Regularization-based methods constrain model behavior to discourage CO. Andriushchenko & Flammarion (2020) proposed maximizing gradient alignment within the perturbation set. Sriramanan et al. (2020) used Euclidean norm regularization while Sriramanan et al. (2021) used nuclear norm regularization to enforce function smoothing in the vicinity of data samples. Lin et al. (2023) identified *abnormal* adversarial examples as a source of CO and mitigated them via regularization. Similarly, Rocamora et al. (2024) encouraged local linearity in the loss surface.

Adaptive Step-Sizes. These methods modify the attack step-size based on training dynamics. Nie et al. (2021) used reinforcement learning to determine step-sizes. Huang et al. (2023) adapted them using gradient norms inspired by Zheng et al. (2020). Zhao et al. (2025) determined the step-size for each sample based on the similarity between the input noise and the gradient direction.

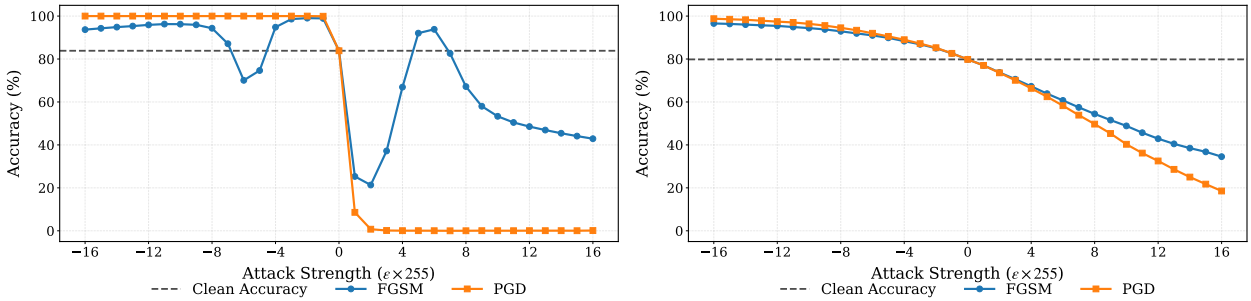
Other Approaches. Some works address CO from alternative perspectives. For instance, Golgooni et al. (2023) omitted updates with negligible gradient magnitudes to stabilize training, while Jia et al. (2022; 2024) use prior-guided adversarial initialization to generate stronger adversarial examples by using perturbations from the historical training process. Zhang et al. (2022); Wang et al. (2025) view FAT from the perspective of bi-level optimization.

2.3. Second-Order Optimization

Second-order methods have been used to enhance adversarial robustness. Inspired by the relation between curvature and robustness (Fawzi et al., 2016), Moosavi-Dezfooli et al. (2019); Ma et al. (2021); Tsiligkaridis & Roberts (2020) train robust models using curvature regularization. Alternatively Singla & Feizi (2020) use second-order information to improve certifiable robustness and Qian et al. (2022) use second-order optimization to generate stronger adversarial examples to improve AT. However, none of these methods fall within the FAT paradigm.

3. Epsilon Overfitting

Although numerous studies have examined CO from various perspectives, a comprehensive understanding of its underlying mechanisms remains elusive. One well-established viewpoint associates CO with changes in the geometry of the model's loss landscape, with several works showing that, at its onset, the loss surface becomes highly distorted (Kim et al., 2021; Kang & Moosavi-Dezfooli, 2021). However, a more precise characterization of this distortion is still lacking. We build on this perspective by analyzing the loss landscape and robustness across different training states, offering a clearer depiction of how the surface evolves throughout CO. Specifically, we investigate whether this distortion follows a consistent, characteristic pattern.



(a) FGSM Adversarial Training

(b) SORA Adversarial Training

Figure 2. Comparison of a model exhibiting CO (a) with a robust model (b), evaluated using FGSM and PGD-10.

3.1. Evolution of Loss Landscape Geometry

A striking property of CO is that it does not degrade clean accuracy, yet it often causes a substantial increase in FGSM accuracy, sometimes even surpassing clean accuracy on the test set. This counter-intuitive effect prompted us to examine FGSM accuracy across a range of ϵ values for a model trained with a fixed perturbation budget (e.g., $\epsilon = 8/255$), as shown in Figure 2(a). See Appendix F.5 for more examples.

Two patterns can be observed in this plot. First, there exist certain ϵ values for which FGSM accuracy drops sharply (e.g., $\epsilon = 2/255$, $\epsilon = -6/255$), while for other values it remains unexpectedly high (e.g., $\epsilon = -2/255$, $\epsilon = 6/255$). Such high variation in FGSM accuracies indicates pronounced nonlinearity in the loss surface, even along the FGSM perturbation direction used during training. Since the model effectively *overfits* to specific ϵ values, performing well for those perturbation magnitudes but poorly for others, we refer to this as *Epsilon Overfitting* (EO).

Epsilon Overfitting arises when the attack perturbation magnitude is fixed and relatively large, with **limited diversity in magnitude or direction**. Under such constraints, the model can learn to reduce the loss only within the narrow regions reached by the fixed perturbation, rather than across the broader ℓ_∞-norm ball. This results in a peculiar form of loss distortion on the onset of CO. See Appendix G for a 3D visualization of the loss surface.

B.S. & Babu (2020); Lin et al. (2023) identified *Abnormal Adversarial Examples* (AAEs), which have lower loss than the original clean samples and become more frequent in the presence of CO. From Figure 1, we see that most AAEs arise in the presence of EO when adversarial examples are generated with the same step-size used during training. In this case, the model enlarges its decision boundary around the adversarial example relative to that around the clean example, producing the observed FGSM accuracy spike, which can even exceed clean accuracy.

3.2. Overcoming Epsilon Overfitting

These findings, along with prior work done by Ding et al. (2020); Huang et al. (2023), highlight the importance of adaptive step-size methods. Such methods adjust step-sizes in response to local loss curvature, avoiding interpolation over highly nonlinear regions between the clean and adversarial points and instead targeting the local maxima along the perturbation direction. Randomization-based methods also try to avoid EO as discussed in Appendix C.1.

The FGSM trend in Figure 2(a), where accuracy evolves in a smooth yet non-monotonic manner for small ϵ, indicates that the loss landscape near the clean point is not purely jagged but exhibits structured curvature. This motivates approximating the local neighborhood around $\epsilon = 0$ with a second-order (quadratic) model, thereby enabling curvature-aware step-size selection for loss maximization.

Although second-order methods can explicitly solve the maximization problem via local curvature estimation, their computational cost is prohibitive for large-scale problems due to Hessian computation (Ghojogh et al., 2021). Consequently, most practical approaches infer curvature indirectly to strike a balance between cost and performance. In the following section, we present a technique for approximating loss surface curvature with **nearly zero** computational overhead.

4. Perturbation Alignment

One of the key characteristics of CO is that, in most cases, it arises very quickly, often within a single epoch. Once CO occurs, most single-step attack methods cannot recover after the robust accuracy drop, and recovery via multi-step evaluation (e.g., PGD) is computationally expensive. Thus, predicting CO **before** single-step and PGD accuracies diverge can enable training methods to take suitable, low-cost corrective actions. Ideally, we seek an accurate and reliable prognosis tool that can detect when optimization has gone off course, well before the onset of CO.

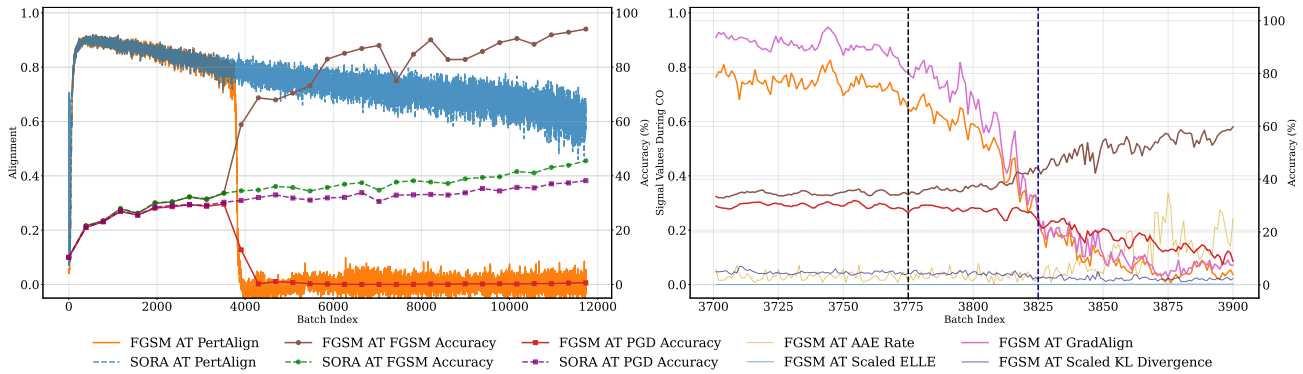


Figure 3. **Left:** Tracking PertAlign during FGSM AT and SORA AT for CIFAR-10 dataset, PertAlign collapses on the occurrence of CO. **Right:** At batch 3775 of FGSM AT, PertAlign and GradAlign begin to drop, forecasting CO, while FGSM and PGD accuracies visibly diverge only around batch 3825. Other metrics, AAE Share of the Batch, Scaled KL Divergence from TRADES, and Scaled ELLE nonlinearity measure, react later in response to the model updates. For more details see Appendix J.

We propose a novel metric, **PertAlign**, which can predict CO by measuring the nonlinearity of the loss surface with negligible additional computational cost. PertAlign is defined as the cosine similarity between:

1. The gradient of the loss with respect to the input, computed during adversarial example generation, and
2. The backpropagation gradient, propagated back to the input layer, computed when performing backpropagation on the adversarial example passed through the model.

Formally **PertAlign** is defined as:

$$\cos\left(\nabla_x \mathcal{L}\big(f_\theta(x'), y\big), \nabla_x \mathcal{L}\big(f_\theta(x'+\delta), y\big)\right), \qquad (2)$$

where $x' = x + \eta$ and $\eta \sim \mathcal{U}(-\epsilon, +\epsilon)^d$ is the random start, and $\delta = \alpha v$ is a perturbation of scalar step-size α in direction v. The perturbation direction v is typically chosen as

$$g = \nabla_x \mathcal{L}\big(f_\theta(x'), y\big) \quad \text{or} \quad p = \operatorname{sign}\left(\nabla_x \mathcal{L}\big(f_\theta(x'), y\big)\right).$$

It can be shown that PertAlign captures the local nonlinearity of the loss surface (see Appendix A).

Lemma 4.1. *Let $f_\theta : \mathbb{R}^d \to \mathbb{R}^C$ be a classifier and let $g = \nabla_x \mathcal{L}(f_\theta(x), y)$, $H = \nabla_x^2 \mathcal{L}(f_\theta(x), y)$. Then for $\alpha \ll 1$,* PertAlign *can be approximated by:*

$$1 - \text{PertAlign} \approx \frac{\alpha^2}{2} \|h_{\perp g}\|^2, \quad h = \frac{Hv}{\|g\|}$$

where $h_{\perp g}$ denotes component of h, orthogonal to g.

More specifically, the misalignment measured by PertAlign correlates with the component of Hv (the Hessian H applied to v) that is orthogonal to the gradient. This misalignment is minimized when the gradient is aligned with Hv, i.e.,

$$v \parallel H^\dagger g, \qquad (3)$$

where $H^\dagger$ is the Moore-Penrose pseudo-inverse of H. This condition closely resembles Newton's method and implies that the perturbation direction should align with the second-order optimal update step in the loss landscape.

In practice, setting $v = p$ yields consistent and reliable behavior, as shown in Figure 3. When CO occurs, PertAlign drops sharply toward zero, indicating that p is no longer aligned with the optimal second-order direction, signaling a failure of the inner maximization step. Conversely, when the loss surface remains approximately linear, PertAlign remains stable rather than collapsing.

As further discussed in Section 3, since at the onset of CO, **before** PGD accuracy declines, the local loss surface becomes highly nonlinear and unstable (Andriushchenko & Flammarion, 2020; Kim et al., 2021; Rocamora et al., 2024), by PertAlign directly measuring this nonlinearity, it can predict CO earlier than the divergence of single-step and PGD accuracies (Figure 3). Moreover, PertAlign consistently foreshadows CO earlier than other proposed metrics.

Note that PertAlign relies only on two quantities: $\nabla_x \mathcal{L}(f_\theta(x), y)$, computed during all single-step methods, and $\nabla_x \mathcal{L}(f_\theta(x+\delta), y)$, the backpropagation gradient extended by one layer to include the input layer, which is already obtained during the model weight update. As these gradients are already part of standard AT, PertAlign incurs **no additional computational overhead**, making it a fast and reliable metric for CO prediction and **can therefore be applied to any other single-step method**. Shafahi et al. (2019); Zhang et al. (2019a) use a similar trick to speed up AT. The information it provides can also be used to mitigate CO, which our proposed AT method leverages in Section 5.

Algorithm 1 **S**econd-**Or**der **A**daptive Method (SORA)

Inputs: # epochs T, # batches M, radius ϵ, step-size α, exponential average coefficient β.

$v \leftarrow 0.99$ {Initialize moving linearity coefficient}
for Epoch $t = 1, \ldots, T$ **do**
  for Batch $i = 1, \ldots, M$ **do**
    $\boldsymbol{\eta} \sim \mathcal{U}(-\epsilon, \epsilon)^d$ {Random start}
    $\boldsymbol{g} \leftarrow \nabla_{\boldsymbol{x}_i} \mathcal{L}\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i + \boldsymbol{\eta}), y_i\right)$
    $\boldsymbol{\alpha}_i \sim \mathcal{U}\left(0, \alpha^*\right)^d$ {Element-wise step sampling}
    $\boldsymbol{x}_i' \leftarrow \boldsymbol{x}_i + \boldsymbol{\eta} + \boldsymbol{\alpha}_i \odot \operatorname{sign}(\boldsymbol{g})$
    $\boldsymbol{x}_i' = \Pi_{[0,1]}(\boldsymbol{x}_i')$ {Project onto the valid pixel range}
    $\boldsymbol{g}', \boldsymbol{g}_{\boldsymbol{\theta}} \leftarrow \nabla_{[\boldsymbol{x}_i', \theta]} \mathcal{L}\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i'), y_i\right)$ {Backpropagation}
    $\boldsymbol{\theta} \leftarrow \text{optimizer}(\boldsymbol{\theta}, \boldsymbol{g}_{\boldsymbol{\theta}})$ {Standard parameters update, (e.g. SGD)}
    Calculate optimal step-size for next batch:

$$\alpha^* \leftarrow \begin{cases} \min\left(\alpha_{\max}, \frac{\alpha_0}{1-v}\right), & v < 1, \\ \alpha_{\max}, & \text{otherwise.} \end{cases}$$

    $v \leftarrow (1-\beta) \cdot v + \beta \cdot \frac{\boldsymbol{p}^T \boldsymbol{g}'}{\|\boldsymbol{g}\|_1}$ {Update moving linearity coefficient}

We further examine related approaches, including detailed PertAlign versus GradAlign (Andriushchenko & Flammarion, 2020) comparison in Appendix C.2, and ZeroGrad and MultiGrad (Golgooni et al., 2023) from a second-order prespective in Appendix C.3. Additional properties of PertAlign, alongside an ablation study, are presented in Appendix A.

5. Methodology

In Section 3, we demonstrated the importance of using an adaptive step-size in mitigating EO and thus CO. This observation motivates us to explore adaptive step-size strategies more systematically. However, this exploration naturally raises a key question:

What is the optimal perturbation step-size, given the current state of the model?

When loss maximization is viewed purely as a first-order optimization problem, our options for determining a suitable step-size are limited. To address this limitation, we instead analyze single-step loss maximization from a second-order optimization perspective.

It can be shown that, by locally approximating the loss landscape with a quadratic function, the optimal step-size for maximizing the loss is given by (see Appendix A for the full derivation):

Lemma 5.1. *For a second order loss function, let the perturbation of the input be in the direction of* $\operatorname{sign}(g)$*:*

$$p = \operatorname{sign}(g) \in \{-1, 0, 1\}^d, \quad v = \alpha p.$$

Then the optimal step-size that maximizes this loss is

$$\alpha^* = \begin{cases} \min\left(\alpha_{\max}, \frac{\alpha_0}{1 - \frac{p^T g'}{\|g\|_1}}\right), & \frac{p^T g'}{\|g\|_1} < 1, \\ \alpha_{\max}, & \textit{otherwise}. \end{cases}$$

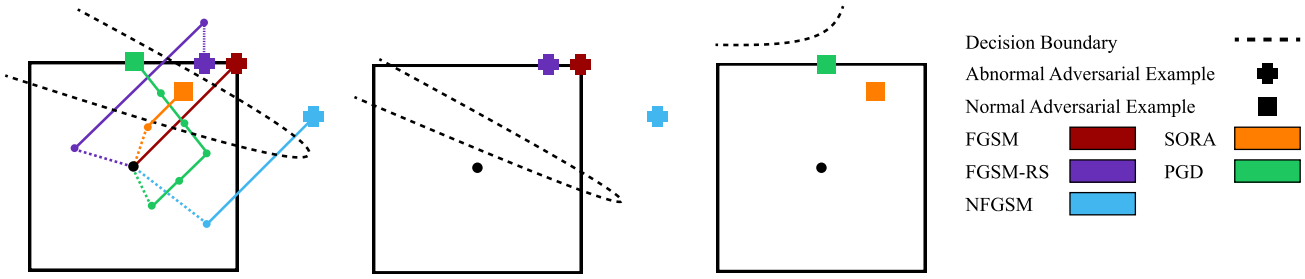
where $g' = g + \alpha H p$*,* $\|\cdot\|_1$ *is the* ℓ_1 *norm, and* $\alpha_{\max}$ *is the maximum step-size budget, and* α_0 *is a fixed numerator.*

Using this formula as a basis for our attack step-size provides a strong initial estimate of an effective perturbation magnitude. To improve stability in cases where the second-order approximation of the loss landscape is inaccurate, and to further mitigate EO, we introduce per-pixel channel diversification: for each channel, we uniformly sample its attack step-size from the range $[0, \alpha^*]$. This stochasticity significantly increases the diversity of perturbations, improving both the robustness and universality of the attack, as demonstrated in Section 6. A summary of our method is provided in Algorithm 1.

Intuitively, SORA adapts to the evolving state of the model by leveraging the backpropagation gradient from the previous batch. By combining this gradient with the general perturbation direction, it estimates the local nonlinearity of the loss surface and adjusts the step-size accordingly.

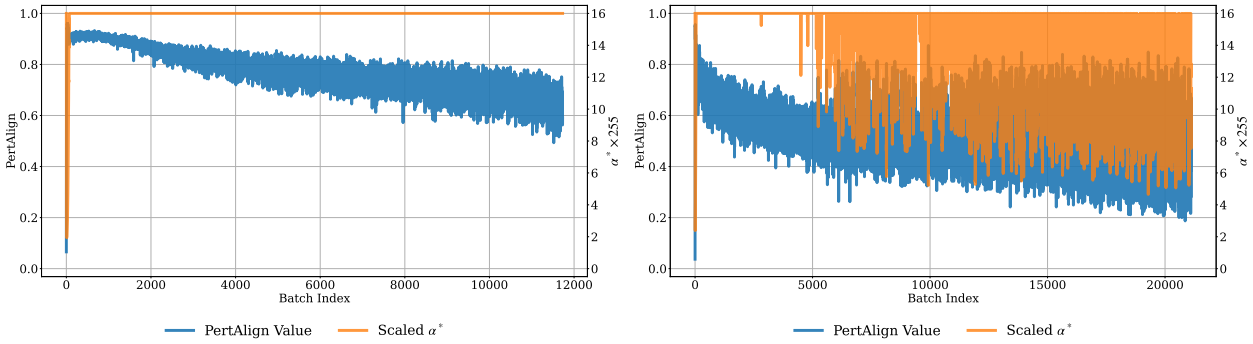
This dynamic adjustment, illustrated in Figure 4, mitigates overshooting, avoids generating AAEs, and prevents the emergence of highly nonlinear regions in the loss landscape. The theoretical derivation of the optimal step-size along the gradient direction is provided in Appendix A.

To match the time complexity of truly fast methods (Wong et al., 2020), we apply our theoretically optimal step size



(a) Before update (b) After update on AAEs (c) After update on NAEs

Figure 4. When the decision boundary becomes distorted, single-step attacks may produce AAEs, whereas multi-step attacks such as PGD can still reliably generate NAEs. By adapting the attack step-size to the local linearity of the loss surface, SORA can also produce NAEs in such scenarios. Training on AAEs tends to exacerbate distortion in the loss surface, while training on NAEs can guide the model toward recovery.



(a) CIFAR-10 (b) PATHMNIST

Figure 5. Theoretically derived and self-adaptive α^* during SORA AT with $\epsilon = 8/255$ on CIFAR-10 (a) and PATHMNIST (b).

with a *temporal gap*. Similar to the approach of Shafahi et al. (2019), which generates adversarial examples with a temporal gap, we use the optimal step-size with a lag, enabling our method to exploit second-order information essentially for *free*. Since the weights change only slightly after each update, we expect the validity of our theoretical foundation to be preserved. We also apply the optimal step-size to a different batch; such extrapolation across batches is not unprecedented in AT (Shafahi et al., 2019; 2020). We believe our theoretical results remain applicable because all batches are drawn from the same training distribution.

The success of our method supports the observation by Wong et al. (2020) that CO is sudden, drastic, and *universal*, compromising all data points, not isolated outliers. Because the problem is global, we adopt a similarly global corrective mechanism. To handle temporal shifts in weights and batches, we stabilize via exponential moving averages as shown in Algorithm 1.

We also tracked the theoretically optimal step-size α^* (Lemma 5.1) during training on CIFAR-10 and PATHMNIST. Figure 5(a) shows that, except for a few early batches, SORA selects the maximum possible step-size throughout training on CIFAR-10, suggesting that the loss landscape is sufficiently linear and does not require step-size reduction to avoid CO. In contrast, on PATHMNIST, SORA consistently readjusts its step-size to achieve robustness (Figure 5(b)). Together with PertAlign's lower average performance on this dataset, this suggests that achieving robustness while avoiding CO is more challenging on PATHMNIST than on CIFAR-10. Thus, failing to incorporate a curvature-aware mechanism in FAT methods may lead to CO on such datasets, as discussed in Section 6.3 and Figure 6. This self-correcting mechanism also enables SORA to perform well with fixed hyperparameters, whereas less flexible methods often require costly setting-specific tuning to avoid CO.

6. Experiments

6.1. Settings

Baselines. We compare our method against a wide range of single-step AT methods, including Free AT (Shafahi et al., 2019), FGSM-RS (Wong et al., 2020), GradAlign (Andriushchenko & Flammarion, 2020), NuAT (Sriramanan et al., 2021), ZeroGrad and MultiGrad (Golgooni et al., 2023), N-FGSM (de Jorge Aranda et al., 2022),

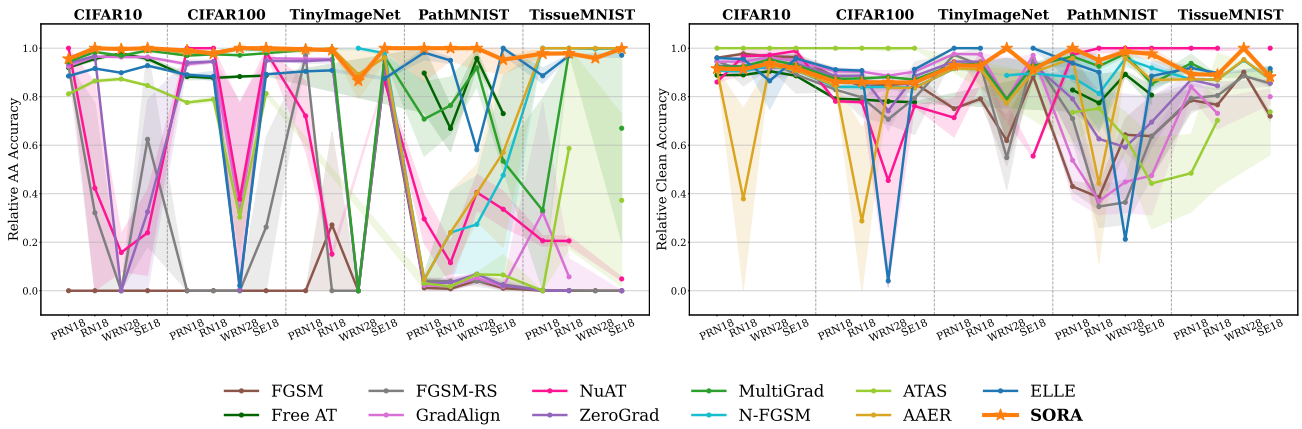
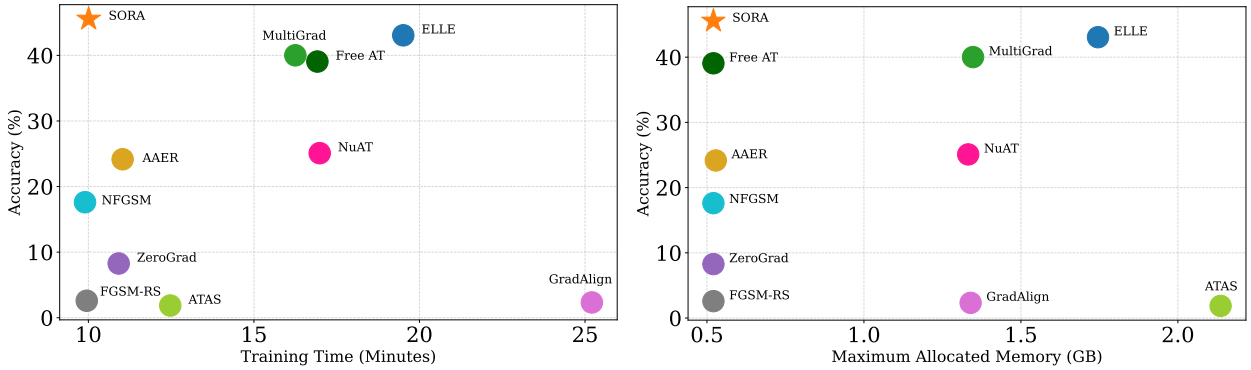


Figure 6. Evaluation of different methods across datasets and architectures. **Left:** SORA attains the highest robust accuracy among single-step methods. **Right:** Corresponding clean accuracy.



(a) Time (b) Memory

Figure 7. Training time vs. memory usage on PATHMNIST with PreActResNet-18, trained for 30 epochs, measured on an NVIDIA GeForce RTX 4090 GPU. The ★ marks SORA. The vertical axis in both figures represents PGD-10 accuracy.

ATAS (Huang et al., 2023), AAER (Lin et al., 2023), and ELLE (Rocamora et al., 2024). We also evaluate multi-step AT methods, including PGD-2 and PGD-10, as well as TRADES (Zhang et al., 2019b), which serve as upper-bound baselines representing idealized performance. We provide more information in Appendix B.

Datasets and Model Architectures. The training datasets include CIFAR-10/100 (Krizhevsky, 2009), and TINY-IMAGENET (Le & Yang, 2015), as well as IMAGENET-100 (Russakovsky et al., 2015). To further evaluate the generalization capability of single-step AT methods, we additionally assess our method and the baselines on PATHMNIST and TISSUEMNIST from the MEDMNIST collection (Yang et al., 2023). More information about these datasets is available in Appendix D. We evaluate on multiple architectures, including ResNet (He et al., 2016a), PreActResNet (He et al., 2016b), and WideResNet (Zagoruyko & Komodakis, 2017) from the ResNet family, as well as SENet (Hu et al., 2018) and ViT (Dosovitskiy et al., 2021). These architectures are discussed in Appendix E.

Hyperparameters. Unless otherwise stated, we set the perturbation budget to $\epsilon = 8/255$ and follow the training setup of Wong et al. (2020), using the SGD optimizer with momentum 0.9 and weight decay 5×10^{-4}. For our method we set $\alpha_0 = 0.02$, $\beta = 0.05$, and $\alpha_{\max} = 2\epsilon$. Exact hyperparameter values and training details are included in Appendices B, D, and E.

6.2. Results

To evaluate the generalizability of single-step AT methods without model- or dataset-specific hyperparameter tuning, we compute each method's AutoAttack (AA) (Croce & Hein, 2020) and clean accuracy relative to the best-performing method in each setting (Figure 6). A relative accuracy closer to 1 indicates performance closer to the state-of-the-art (SOTA) in that setting. For comprehensive results, see Appendix K where we also reported AA for all baselines.

Table 1. Training results for PreActResNet-18 on PATHMNIST and IMAGENET-100 for $\epsilon = 8/255$. The performance gap of each method with respect to SORA is also shown in the parentheses.

Method	PATHMNIST		IMAGENET-100	
	Clean	AutoAttack	Clean	AutoAttack
FGSM (Goodfellow et al., 2015)	46.06 (-39.87%)	0.42 (-32.49%)	15.94 (-41.32%)	0.00 (-18.56%)
N-FGSM (de Jorge Aranda et al., 2022)	74.89 (-11.04%)	1.65 (-31.26%)	49.40 (-7.86%)	15.52 (-3.04%)
AAER (Lin et al., 2023)	83.54 (-2.39%)	2.48 (-30.43%)	48.26 (-9.00%)	17.18 (-1.38%)
SORA (Ours)	**85.93%**	**32.91%**	**57.26%**	**18.56%**

SORA is the only fast method that consistently matches or surpasses the SOTA across all settings, while maintaining higher clean accuracy than competing methods with comparable robust accuracy, and minimal computational overhead (Figure 7). This demonstrates a superior trade-off between robustness, clean accuracy and efficiency.

SORA's effectiveness becomes even more pronounced on more challenging datasets with finer image textures and higher resolution, where it consistently outperforms strong baselines such as N-FGSM and AAER in terms of both clean and robust accuracy, as shown in Table 1.

6.3. Discussion on Universality

Figure 6 reveals a pronounced performance gap between prior methods on both standard datasets (CIFAR-10/100, TINYIMAGENET) and more challenging datasets (PATHMNIST, TISSUEMNIST). This highlights a **lack of generalization** in previous approaches (see Appendix H).

Robustness on PATHMNIST and TISSUEMNIST is particularly challenging due to finer image textures, shorter inter-class distances compared to datasets such as CIFAR-10 (Figure 9), and class imbalance (Figure 8). Many methods that avoid CO and perform well on standard benchmarks suffer noticeable drops in robust accuracy on these harder datasets. Furthermore, approaches that do maintain robustness, such as ELLE and MultiGrad, incur substantially higher computational or memory costs (Figure 7) or sacrifice clean accuracy such as N-FGSM (Table 1).

Even when we conduct hyperparameter search for these baselines, we are unable to find any configuration which performs well on PATHMNIST or even when such a configuration exists, it results in a significant performance drop on other datasets such as CIFAR-10, indicating the lack of a universal set of hyperparameters necessary for a FAT method. For more details see Appendices H and M.3.

6.4. Contribution of SORA's Components

We perform an ablation study to evaluate the contribution of each component of SORA to its overall performance. Experiments are conducted using PreActResNet-18 on the PATHMNIST dataset, which is more susceptible to CO than CIFAR-10/100, and thus better exposes the effect of each component. As shown in Table 2, increasing the variability of the step-size magnitude significantly improves robust accuracy; most notably, removing the optimal step-size selection results in the lowest PGD-10 accuracy.

Table 2. Ablation study on the components of SORA.

Configuration	Clean	FGSM	PGD-10
SORA (baseline)	84.69	57.51	45.56
– Without Random Sampling	85.67	58.89	45.35
– Clamping Step-Size	86.82	52.02	34.08
– Without Optimal Step-Size	89.93	28.30	17.23

These results highlight the importance of both attack strength and directional variability, as discussed in Section 3, and demonstrate the necessity of adaptive step-size selection for challenging datasets.

7. Conclusion

We analyzed loss surface distortion in CO, revealing the critical role of the attack perturbation distribution in Epsilon Overfitting (EO) and its close connection to CO. We further introduced PertAlign, a novel, cost-free metric for reliably predicting CO based on loss surface curvature. Building on insights from EO and PertAlign, we developed SORA, an adaptive step-size method that dynamically adjusts its perturbation distribution according to local linearity.

Extensive experiments across 15 baselines, 6 datasets, and 4 architectures demonstrate that SORA consistently prevents CO, achieves SOTA performance, and generalizes robustly across datasets and architectures, settings in which many existing methods fail. We believe that PertAlign, together with the criteria highlighted in this work, provides a principled foundation for developing improved fast adversarial training methods in future research, enabling better generalization and more reliable evaluation through broader and more diverse experimental validation.

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Impact Statement

This paper presents work whose goal is to advance the field of trustworthy machine learning by solving a key practicality problem in adversarial robustness. The high computational cost of robust training is a major barrier to its widespread adoption, leaving many real-world systems vulnerable. We show that many existing *fast* methods hide large tuning costs, which are frequently a necessary response to instabilities like **Catastrophic Overfitting**. To address this issue, we propose **SORA**, a method that offers true training efficiency and stability. By significantly lowering the barrier to obtaining robust models, this work could help secure vital ML applications. We consciously acknowledge that efficient training still consumes energy, and that improved robustness, while critical for security, is one component of a broader effort needed to build safe and responsible AI.

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